Projects in Mathematics and Applications

BULK RNA-SEQ QUANTIFICATION

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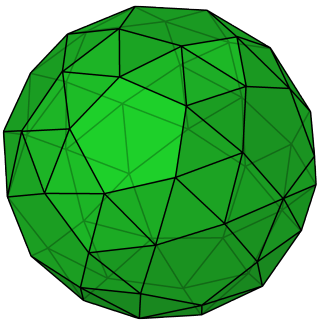
July 2023



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Acknowledgments

First and foremost, we would like to express our sincere gratitude to our mentors, Mr. Nguyen Trung Nghia and Mr. Tran Hoang Bao Linh, for their dedicated guidance throughout our learning process and the completion of this report.
We would like to extend our heartfelt thanks to the PiMA 2023 organizing committee for providing lectures, knowledge, and contributing to the facilities that have supported the completion of our report.
Special thanks to Mr. Vuong Hy from BioTuring for his valuable feedback and corrections on our report.
Due to the limitations of our knowledge and theoretical abilities, there are undoubtedly shortcomings and constraints in this work. We sincerely look forward to the guidance and contributions from our respected teachers to further enhance the quality of our report.
Once again, thank you very much!

Summary

The report provides a brief overview of Bulk RNA-Seq Quantification technology and the associated algorithmic models.

The application of the next-generation sequencing technology, RNA-Seq, has become more widespread than ever for studying the transcriptome and conducting drug experiments. Used in the early 2000s, studies on prostate cancer cells (2006), the flowering plant Medicago truncatula (2006), and corn (2007) became the first to utilize RNA-Seq. The technology has gained even more attention following the implementation of the Encyclopedia of DNA Elements (ENCODE) project and The Cancer Genome Atlas (TCGA) project.

The RNA-Seq technology is divided into two types: Bulk RNA-Seq and Single-cell RNA-Seq. The difference between these two methods is that bulk RNA-Seq uses samples containing multiple cells, while single-cell RNA-Seq uses one cell per sample. This report will primarily focus on bulk RNA-Seq and elaborate on the related algorithms.

In this report, we will employ Bayesian graphs and the EM (Expectation–Maximization) technique to address the problem.

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1 RNA-SEQ OVERVIEW

1.1 Introduction

The genome is a collection of genes present in a living system. Studying the genome helps us understand more about ourselves and how genes interact with each other as well as the surrounding environment. However, to truly understand the function of genes and how they operate, we need to study their products, which are RNA and proteins. Research on the transcriptome (collection of RNA in a living system) and proteome (collection of proteins in a living system) is rapidly advancing with the breakthrough of next-generation sequencer technology.

In this article, we will primarily focus on the study of gene expression through transcripts in the transcriptome using bulk RNA-Seq technology. RNA-Seq, also known as RNA Sequencing, was created and began to be used in the early 2000s. This technology quantifies specific RNA amounts, detects various types of pre-mRNA splicing (including introns and exons), and provides an overview of the transcriptome.
RNA-Seq technology is divided into two types: bulk RNA-Seq and single-cell RNA-Seq. Bulk RNA-Seq analyzes multiple cells at once, resulting in the average of the cells. In contrast, single-cell RNA-Seq analyzes individual cells, with each cell producing a different result. Therefore, bulk RNA-Seq usually requires 6 samples (each sample including multiple cells), while single-cell RNA-Seq may require over 800 samples (each sample including only one cell). This leads to higher costs when using single-cell RNA-Seq methods.
In addition to RNA-Seq, microarray technology is also used to analyze the transcriptome. However, the main drawback of microarrays compared to RNA-Seq is the inability to detect new RNAs and difficulty in determining low-expressed genes.

1.2 Bulk RNA-Seq Process

1.2.1 *Sample Preparation*

The first step is to collect samples (blood, skin, etc.), with each sample containing multiple cells. After obtaining a sufficient number of large samples, the next step is to extract RNA from the cells. This can be done using oligo-dT or rRNA depletion methods. Since the focus of this report is mRNA, we will discuss the use of the oligo-dT method. Oligo-dT relies on the poly-A tail of mRNA, a chain of adenine nucleotides attached at the end of mRNA. By binding to the poly-A tail, particles containing oligo-dT can slowly separate mRNA from the cells. Then, we can select specific mRNA types for analysis.
It is important to note that during the mRNA separation process, some mRNA may be lost, especially mRNAs shorter than 100 base pairs (bp).
Next, isolated mRNA is fragmented into small pieces called fragments (usually ranging from 100 to 300 bp in length). Various methods, such as physical methods (e.g., sonication), enzymatic cutting (e.g., RNAse II, transposase), or chemical methods (e.g., temperature), can be used for the fragmentation process. After obtaining the mRNA fragments, which are small cut molecules of mRNA, we need to convert them into a more stable structure than mRNA, which is DNA. The reason for this is that mRNA is relatively unstable and can easily degrade within 14 to 24 hours. This conversion process is called cDNA synthesis (cDNA: complementary DNA). We initiate this process by using the mRNA fragments as a

template to transcribe into a new DNA strand, following the principle of complementarity (A pairs with T, G pairs with C). This synthesis is catalyzed by an enzyme called reverse transcriptase, which is used by retroviruses like HIV to transcribe their RNA into DNA and insert it into the host cell's genome.
Next, we synthesize a second DNA strand complementary to the first one using the enzyme DNA polymerase, a DNA replicating enzyme. After completing this process, we obtain a complete cDNA consisting of two DNA strands. At this point, the fragments are considered cDNA and are no longer mRNA. However, a challenge arises because we cannot determine which mRNA the synthesized cDNA corresponds to. For example, if we have two mRNA strands with complementary sequences (from transcription of two genes on two DNA strands with complementary sequences), and we use each mRNA strand as a template to transcribe into a first DNA strand, we end up synthesizing a second DNA strand from the first one. In the end, we obtain two identical cDNA molecules, even though they were transcribed from two different mRNA molecules with distinct sequences and functions.
To address this issue and avoid ambiguity in identifying the mRNA from which cDNA is synthesized, dUTP (uracil instead of thymine) can be used in the second DNA synthesis step. This modification involves replacing thymine (T) with uracil (U) in the synthesized DNA strand. After converting mRNA fragments to cDNA, we need to attach adapters to both ends of the cDNA. Adapters are nucleotide sequences that can bind to the ends of DNA or RNA strands. The structure of an adapter includes three parts: P5/P7, ID, and the adapter sequence. P5/P7 contains sequences that allow cDNA to bind to specific positions in the sequencing machine. ID contains sequence lines that help mark and differentiate between different cDNAs. The adapter sequence is the nucleotide sequence that helps primers bind for DNA amplification.
During the adapter attachment process, some cDNAs may not get attached, leading to their exclusion from subsequent processes. To increase the quantity of the current cDNA fragments for analysis and reduce errors, the Polymerase Chain Reaction (PCR) technique is employed. PCR is a technique that produces numerous copies of the original DNA fragments outside a living organism. In the context of this report, specific details about the PCR process are not covered, but it can be understood as similar to the DNA replication process inside cells.
Before using PCR, uracil DNA glycosylase (UDG) is applied to remove uracil nucleotides from the DNA strand containing them. This process renders those DNA strands incomplete due to the loss of uracil nucleotides, making it difficult to replicate them efficiently. Moreover, strands of DNA that did not attach adapters during the previous steps will not be replicated during PCR.
Once a sufficient number of copies of the initial cDNA are obtained, the next step is to assess the quality of the existing cDNA. This quality assessment involves measuring library concentration and fragment length. Library concentration determines the quantity of cDNA in the sample, with an average of 20-30 million fragments per sample recommended. The selection of fragments with similar lengths, typically ranging from 150 to 200 base pairs, is crucial for the research purpose.

1.2.2 *Sequence decoding*

After sample preparation, the sequencing of fragments in the sample is performed. At this stage, the fragments become reads, meaning they have been sequenced. Once the reads are completed, a FASTQ file is generated, containing the nucleotide sequence, ID, and reading quality of all the reads. Reading quality is a scale from 1 to 10 that evaluates the sequencing

quality of the machine. During the reading process, errors such as misidentification of a nucleotide or misreading may occur, resulting in reads that are not accurately sequenced and, therefore, cannot be included in the analysis. Hence, reads with a reading quality below 6 are usually removed from the dataset.

Sequencing helps us determine which gene in the genome the read belongs to and allows us to count the number of reads associated with that gene. It's important to note that the procedures described above are carried out for Illumina short-read machines capable of sequencing. Illumina is a machine designed specifically for sequencing with a large number of short fragments, typically ranging from 50 to 500 base pairs (10^9 to 10^10 reads per run). Due to its ability to generate large datasets with good quality, cost-effectiveness, and user-friendliness, Illumina is commonly used for analyzing differential gene expression.

In addition to Illumina, other machines such as Pacific Biosciences (PacBio) long-read and Oxford Nanopore (ONT) direct RNA-sequencing are used. These machines can sequence an entire transcript without the need to fragment it into pieces like Illumina. PacBio can sequence an entire cDNA synthesized from mRNA, eliminating the need for the fragmentation step in sample preparation. ONT can sequence mRNA directly without converting it to cDNA, avoiding the need for fragment creation or the cDNA synthesis process. However, the major drawback of these two machines is a higher error rate in reading as well as a lower throughput (10^6 to 10^7 reads in one run) compared to Illumina, with higher associated costs. On the other hand, because they can sequence the full length of a transcript, these machines allow for the discovery of new transcript types or the analysis of specific mutations on a transcript.

The next step involves data analysis and includes a total of four stages. However, depending on the specific purpose of performing RNA-Seq, these stages may vary, but the fundamental principles are common.

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Stage 1 - Alignment sequencing read: After obtaining the FASTQ file containing the sequence of reads, the first step is to compare these reads with the genome (or transcriptome). This helps determine one or more genomic coordinates (locations on the gene map) for each read. Several tools are used for this stage, categorized into those comparing reads with the genome (e.g., TopHat, STAR, HISAT), those comparing reads with the transcriptome (e.g., StringTie, SOAPdenovo-Trans), and those capable of both stages (e.g., Sailfish, Kallisto, Salmon).

Stage 2 - Quantification of transcript abundance: Once genomic coordinates of reads are determined, indicating their positions on the genome or transcriptome, this stage involves identifying which gene/transcript they belong to. This stage significantly influences the results throughout the data analysis process. The challenge lies in the fact that a read may have multiple genomic coordinates due to the existence of multiple transcripts/genes with overlapping sequences. Several tools are developed for Stage 2, with different approaches to address this challenge. One commonly used tool is HTSeq, which often excludes reads with multiple genomic coordinates, potentially excluding transcripts with overlapping sequences from the results. Additionally, there is RSEM, a popular tool that uses the expectation maximization algorithm to handle reads with sequences overlapping on multiple transcripts/genes. The EM algorithm is a central focus of this report and will be explained in detail in the "Building the Bulk RNA-Seq Quantification Problem" section. After quantifying the transcripts, the outcome of this stage is a gene expression matrix containing the number of reads or estimated quantities for each gene/transcript.

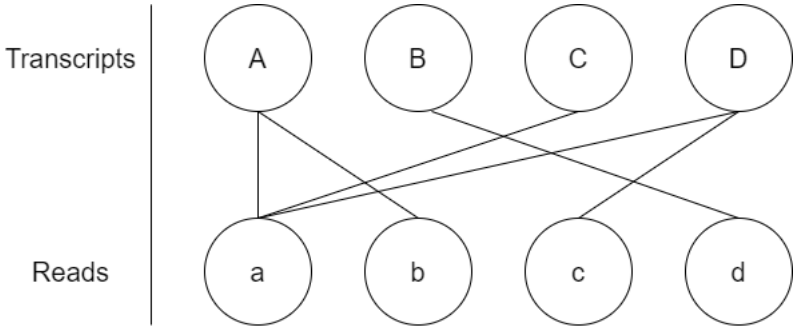
Stage 3 - Filtering and normalization: After obtaining the gene expression matrix, the next step is to filter and normalize it to draw accurate biological conclusions. During this stage, factors such as differences between samples, technical errors, and outliers need to be considered.
Stage 4 - Differential expression modeling: The final stage of the data analysis process involves building models based on the gene expression matrix. Tools like edgeR or CuffDiff are used to determine changes in gene expression through transcripts.

2 Constructing the Bulk RNA-Seq Quantification Problem

2.1 Problem Overview

Our task is to quantify the amount of mRNA for each transcript type based on:

- A given transcriptome
- A set of reads, each with a specified quantity
- The ability of a transcript to produce reads upon mapping



At this point, we encounter challenging issues

- Multiple reads may map to a single transcript, and vice versa.
- Reads may share identical sequences.

2.2 Problem-Solving Approach

During the execution of Bulk RNA-Seq, probabilistic factors affect the obtained read set (start position of fragment cutting, length of cut fragments, etc.). Transforming these factors into random variables and setting parameters, we utilize a Bayesian network model to visually represent these influences.
With the Maximum Likelihood Estimation (MLE) method, we can construct a likelihood function and find the maximum using Lagrange Multiplier. However, the resulting system is not easily solvable. Instead, employing the Expectation-Maximization (EM) algorithm with a set of sample parameters to maximize the expectation function via Lagrange Multiplier yields a simpler system for computation.

2.3 Symbol Table

Symbols	Meaning
Ω	Parameter Set in Bayesian Graph
Ω_{old}	Sample Parameter Set
$Ar = a$	Set of Alignments a is a set (t, f, s) uniformly sampled from a specific read
(t_a, f_a, s_a)	A specific set corresponding to a specific alignment a
A	Set containing all elements of Ar
r	Read
t	Transcript
l	Length of transcript
f	Length of fragment
s	Starting position of fragment
$P(X\|Y)$	Probability mass function of X given Y
$\mathbb{E}(X)$	Expectation of continuous random variable X
$\forall x$	For all x
X	Set X containing elements x
$x \in X$	x is an element of set X
$\sum_{n \in N} f(n)$	Sum of $f(n)$ over n in set N
$\prod_{n \in N} f(n)$	Product of $f(n)$ over n in set N
$\frac{\partial f}{\partial x}$	Partial derivative of f with respect to x
$x^* = \text{argmax}_x f(x)$	Value x at which function $f(x)$ achieves its maximum
Cat	Categorical distribution
MLE	Maximum Likelihood Estimation
Lagrange Multiplier	Lagrange Multiplier method
EM algorithm	Expectation-Maximization algorithm

2.4 Generative model

1. Select a specific transcript t_i from the transcriptome set with a probability of θ_{t_i} ($t \sim \text{Cat}(\theta)$).

2. Choose the fragment length f_i with parameter λ adjusted based on the length of transcript t_i ($f \sim \text{Cat}(\lambda)$).

3. Choose the starting point for preparation of the cut, denoted as s_i, with a length of f_i (in this report, we assume $s \sim$ Uniform).

4. Cut out a fragment from s_i to $s_i + f_i - 1$.

5. Read the sequence r as a portion of that fragment.

3 Expectation-Maximization Algorithm (EM algorithm)

3.1 Bayesian Network

Introduction: Bayesian Network is a representation in the form of a directed acyclic graph of statistical dependencies among a set of random variables, where:

- Nodes represent variables.
- Edges represent conditional dependencies.

The joint probability distribution of variables is determined by the graph structure of the network and generated according to the formula:

$$P(X_1, X_2, ...) = \prod_{\forall X_i} P\left(X_i | \text{parents}(X_i)\right)$$

With X_i as variables, parents(X_i) as variables dependent on X_i. Building a Bayesian network leads to models that are easy to interpret and efficient inference algorithms.

3.2 Lagrange Multiplier

Introduction: Lagrange Multiplier is a method to find the local minimum or maximum of a function subject to constrained conditions.
Problem: Given a function $f(x)$ subject to $g(x) = 0$, we need to find the minimum or maximum of $f(x)$.
Set up the Lagrange function: $\mathcal{L}(x, \lambda) \equiv f(x) + \lambda \cdot g(x)$
Find the stationary point of $\mathcal{L}$, which means $\dfrac{\partial \mathcal{L}}{\partial x} = 0$ and $\dfrac{\partial \mathcal{L}}{\partial \lambda} = 0$ (1)
Equivalently, $\dfrac{\partial f(x)}{\partial x} + \lambda \cdot \dfrac{\partial g(x)}{\partial x} = 0$ and $g(x) = 0$.

The above method is suitable for cases where the system of equations in (1) is solvable directly, providing a solution without the need for multiple iterations. However, it is easy to observe that the system of equations obtained in (1) is not always easy to solve.

3.3 Expectation-Maximization Algorithm (EM algorithm)

Introduction: The EM algorithm is an iterative method to obtain a maximum likelihood estimate (MLE) of parameters in a statistical model when there are hidden (latent) variables. This method iterates through 2 steps:
- *E-step* generates an expectation function for hidden data and the log-likelihood function conditioned on obtaining a sample parameter set Ω_{old}.
- *M-step* maximizes the expectation value, updating to a new set of parameters Ω_{old}.
The iteration stops when a new parameter is obtained that is not significantly different from Ω_{old} (convergence).

Algorithm Construction:

Input: Consider a statistical model consisting of set R containing observed data variables and set A containing variables with hidden data.
Initialization: Choose the parameter set Ω_{old} for the distribution $P(R, A, \Omega)$
While not converged **do** continue updating EM
- *E-step:* Calculate the posterior distribution $P(R, A|\Omega)$ and the expectation function

$$Q(\Omega|\Omega_{old}) = \mathbb{E}_{A|R,\Omega_{old}}[\log P(R, A|\Omega)]$$

- *M-step:* Maximize the expectation, updating Ω_{old}

$$\Omega_{old} = \underset{\Omega}{\operatorname{argmax}}\, Q(\Omega|\Omega_{old})$$

4 Bulk RNA-Seq Quantification Level 1

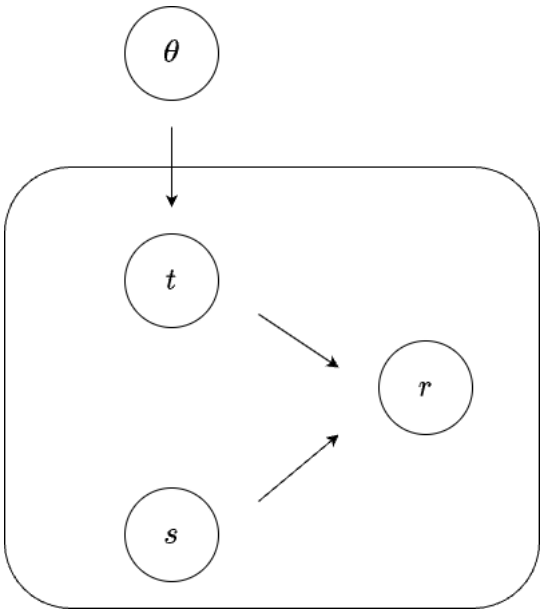
4.1 Assumptions

- Transcripts have equal abundance, denoted by l.
- Fragments are considered as reads.
- Reads have equal length, denoted by f.

4.2 Constructing Bayesian Network

We build a Bayesian network based on the following relationships:

- $\theta \rightarrow t$: Probability that t_i is randomly chosen from the transcriptome is θ_i.
- $t \rightarrow r$: The selected transcript for cutting and analysis influences the generation of fragments and determines a specific read. For example, Transcript A will produce fragments/reads a_n, and Transcript B will produce fragment/read b_n. If Transcript A is chosen for analysis, read b_n cannot appear.
- $s \rightarrow r$: The starting position for cutting and creating a read with length f also influences the type of read. For example, for a transcript sequence AUGGGCUA and $f = 3$ nucleotides, if $s_i = 3$, the generated read is GGG. If $s_i = 5$, the generated read is GCU.



Building and optimizing the likelihood function
Joint probability formula
Using Bayes' Theorem and the Bayesian network constructed above, we have:

$$P(r,t,s|\theta) = P(t|\theta)P(s|t,\theta)P(r|t,s,\theta) = P(r|t,s)P(s)P(t|\theta)$$

Likelihood function
Define the likelihood function as follows: Let $L(\Omega|R)$ be the likelihood function.

$$\begin{aligned}
L(\Omega|R) &= P(R|\Omega) \\
&= \prod_{r\in R} P(r|\Omega) \\
&= \prod_{r\in R}\left(\sum_{t_i} P(r|t_i)\times\theta_{t_i}\right) \\
&= \prod_{r\in R}\left(\sum_{t_i}\sum_{s} P(r|t_i,s_i)\times P(s_i)\times\theta_{t_i}\right) \\
&= \prod_{r\in R}\left(\sum_{(t_i,s_i)} P(r|t_i,s_i)\times\theta_{t_i}\times\frac{1}{l-f+1}\right)
\end{aligned}$$

(The first equality is by definition, the second is due to the independence of reads r, the third separates Ω into θ_{t_i}, the fourth considers cases of s, and the fifth assumes the length of the transcript is l, so there are $l-f+1$ cutting positions, assuming equal probabilities)

4.3 Computing EM Updates

E-step: Expectation function

$$\begin{aligned}
Q(\Omega|\Omega_{old}) &= \mathbb{E}_{A|R,\Omega_{old}}[\log P(R,A|\Omega)] \\
&= \sum_{r\in R}\mathbb{E}_{a|r,\Omega_{old}}[\log P(r,a|\Omega)] \\
&= \sum_{r\in R}\sum_{a\in Ar}\log(P(r,a|\Omega)).P(a|r,\Omega_{old})
\end{aligned}$$

M-step:
Transcript distribution
Using the Lagrange Multiplier method, taking partial derivatives of $Q(\Omega|\Omega_{old})$ with respect to θ_{t^*} and assuming they are equal to β, we can solve these equations with a constraint that the sum of each variable is 1.
Constraint $\sum_{\forall t_i} \theta_{t_i} = 1$
Consider

$$P(r, a|\Omega) = P(t_a|\Omega).P(r, a|t_a, \Omega)$$

Fixing a t^* in T, we observe that the probability $P(r, a|t_a, \Omega)$ is independent of θ, and taking the derivative with respect to θ_{t^*} we get

$$\begin{aligned}
\frac{\partial Q(\Omega|\Omega_{old})}{\partial \theta_{t*}} &= \sum_{r \in R} \sum_{a \in Ar} \frac{\partial \log[\theta_{t_a}.P(r, a|t_a, \Omega)]}{\partial \theta_{t^*}}.P(a|r, \Omega_{old}) \\
&= \sum_{r \in R} \sum_{a \in Ar} \frac{\partial \log(\theta_{t_a})}{\partial \theta_{t^*}}.P(a|r, \Omega_{old}) \\
&= \sum_{r \in R} \sum_{a \in Ar, t_a = t^*} \frac{1}{\theta_{t^*}}.P(a|r, \Omega_{old})
\end{aligned}$$

From here, we get a system of equations

$$\sum_{r \in R} \sum_{a \in Ar, t_a = t^*} \frac{1}{\theta_{t^*}}.P(a|r, \Omega_{old}) = \beta$$

$$\sum_{\forall t_i} \theta_{t_i} = 1$$

Based on the variable β, we can derive

$$\theta_{t^*} = \frac{1}{\beta}. \sum_{r \in R} \sum_{a \in Ar, t_a = t^*} P(a|r, \Omega_{old})$$

Considering the sum of all θ_t, we can solve for β

$$\sum_{\forall t_i} \theta_{t_i} = \frac{1}{\beta}. \sum_{r \in R} \sum_{a \in Ar} P(a|r, \Omega_{old}) = \frac{1}{\beta}. \sum_{r \in R} 1 = \frac{|R|}{\beta}$$

$$\frac{|R|}{\beta} = 1$$

$$\beta = |R|$$

Finally, we have

$$\theta_{t^*} = \frac{1}{|R|}. \sum_{r \in R} \sum_{a \in Ar, t_a = t^*} P(a|r, \Omega_{old})$$

5 Bulk RNA-Seq Quantification Level 2

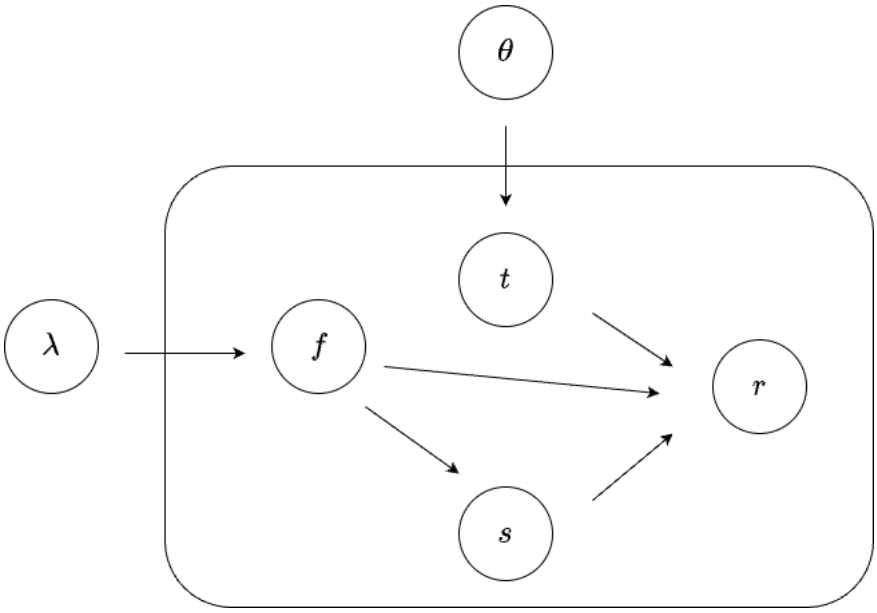
5.1 Assumptions

- Transcripts have equal abundance, denoted by l.
- Fragments have lengths $f_i \sim \text{Cat}(\lambda)$ that vary.

5.2 Constructing Bayesian Network

In addition to the connections at level 1, we have new connections created:

- $\lambda \to f$: By definition, f depends on λ.
- $f \to s$: The length of the read will influence the starting position of the read. For example, for a transcript AAUU and a read length of 3, the starting position can only be 1 or 2.
- $f \to r$: The length of the read cut will influence the type of read. For example, for a transcript with the sequence AUGGGCUA and $s_i = 3$, if $f_i = 3$, the read is GGG. If $f_i = 5$, the read is GGGCU.



Building and optimizing the likelihood function
Joint probability formula
Built from Bayes' formula and Bayesian network as in level 1, similarly, we have

$$P(r, t, f, s|\theta, \lambda) = P(r|t, f, s)P(t|\theta)P(s|f)P(f|\lambda).$$

Likelihood function
We also define the likelihood function as in Level 1, with (Ω now including θ_{t_i} and λ)

$$\begin{aligned} L(\Omega|R) &= P(R|\Omega) \\ &= \prod_{r \in R} \left(\sum_{t_i, f_i, s_i} P(r, t_i, f_i, s_i|\Omega) \right) \end{aligned}$$

5.3 Computing EM Updates

E-step: Expectation function

$$\begin{aligned} Q(\Omega|\Omega_{old}) &= \mathbb{E}_{A|R,\Omega_{old}}[\log P(R,A|\Omega)] \\ &= \sum_{r\in R} \mathbb{E}_{a|r,\Omega_{old}}[\log P(r,a|\Omega)] \\ &= \sum_{r\in R}\sum_{a\in Ar} \log(P(r,a|\Omega)).P(a|r,\Omega_{old}) \end{aligned}$$

M-step:
Transcript distribution
Based on the joint probability formula, we have

$$P(r,a|\Omega) = P(r|a)P(t_a|\theta)P(s_a|f_a)P(f_a|\lambda) = P(t_a|\theta).R$$

Fixing a t^* in T, we observe that the product of probabilities R is independent of θ, and taking the derivative with respect to θ_{t^*} we get

$$\begin{aligned} \frac{\partial Q(\Omega|\Omega_{old})}{\partial \theta_{t^*}} &= \sum_{r\in R}\sum_{a\in Ar} \frac{\partial \log(\theta_{t_a}.R)}{\partial \theta_t}.P(a|r,\Omega_{old}) \\ &= \sum_{r\in R}\sum_{a\in Ar} \frac{\partial \log(\theta_{t_a})}{\partial \theta_{t^*}}.P(a|r,\Omega_{old}) \\ &= \sum_{r\in R}\sum_{a\in Ar,t_a=t^*} \frac{1}{\theta_{t^*}}.P(a|r,\Omega_{old}) \end{aligned}$$

With the derivative result similar to level 1, by the same method, we solve for

$$\theta_{t^*} = \frac{1}{|R|}.\sum_{r\in R}\sum_{a\in Ar,t_a=t^*} P(a|r,\Omega_{old})$$

Fragment length distribution
According to the joint probability formula, we have

$$P(r,a|\Omega) = P(r|a)P(t_a|\theta)P(s_a|f_a)P(f_a|\lambda).$$

Fixing an f^* in F, we observe that except for $P(f_a|\lambda)$, the other probabilities above are independent of λ.
On the other hand, we have

$$P(f_a|\lambda) = \lambda_{f_a}$$

Taking the derivative with respect to λ_f we get

$$\begin{aligned} \frac{\partial Q(\Omega|\Omega_{old})}{\partial \lambda_{f^*}} &= \sum_{r\in R}\sum_{a\in Ar,|f_a|\le l} \frac{\partial \log(\lambda_{f_a})}{\partial \lambda_{f^*}}.P(a|r,\Omega_{old}) \\ &= \sum_{r\in R}\sum_{a\in Ar,|f_a|\le l} \frac{1}{\lambda_{f^*}}.P(a|r,\Omega_{old}) \end{aligned}$$

Using the Lagrange Multiplier method similar to the transcript distribution, we solve for

$$\lambda_f = \frac{1}{|R|}.\sum_{r\in R}\sum_{a\in A_r,|f_i|=f} P(a|r,\Omega_{old})$$

6 Conclusion, Evaluation, and Future Directions

In the model at level 3, with the Bayesian graph and likelihood function already calculated, and since there are only parameters θ and λ, the updating process is somewhat similar to the model at level 2. It will update each parameter, and the crucial point is that the derivative calculations for each variable in the EM update will differ.
The mathematical model at all three levels still lacks the inclusion of errors. These errors may arise from technical issues during sample collection or RNA sequencing. Therefore, the development direction in the future for the project could involve incorporating these errors.

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